CHAPTER 6

Orthogonality and Least Squares

（正交性和最小二乘）

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Chapter 6 Orthogonality and Least Squares

- § 6.1 Inner Product, Length and Orthogonality
- § 6.2 Orthogonal Sets
- § 6.3 Orthogonal Projections
- § 6.4 The Gram – Schmidt Process
- § 6.5 Least – Squares Problems

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§ 6.1 Inner Product, Length and Orthogonality

- 1. The Inner Product
- 2. The Length of a Vector
- 3. Distance in R^n
- 4. Orthogonal Vector
- 5. Orthogonal Complements

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1. The Inner Product(内积)

Inner product or **dot product** of

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} :$$

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T\mathbf{v} = \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = u_1v_1 + u_2v_2 + \cdots + u_nv_n$$

$$\mathbf{v} \cdot \mathbf{u} = v_1u_1 + v_2u_2 + \cdots + v_nu_n = u_1v_1 + u_2v_2 + \cdots + u_nv_n = \mathbf{u} \cdot \mathbf{v}$$

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- Example：$\boldsymbol{u}=\begin{bmatrix}2\\-5\\-1\end{bmatrix}$，$\boldsymbol{v}=\begin{bmatrix}3\\2\\-3\end{bmatrix}$，计算 $\boldsymbol{u}\cdot\boldsymbol{v}$ 和 $\boldsymbol{v}\cdot\boldsymbol{u}$.

- Solution：

$$\boldsymbol{u}\cdot\boldsymbol{v}=\boldsymbol{u}^{\mathrm{T}}\boldsymbol{v}=\begin{bmatrix}2 & -5 & -1\end{bmatrix}\begin{bmatrix}3\\2\\-3\end{bmatrix}=(2)(3)+(-5)(2)+(-1)(-3)=-1$$

$$\boldsymbol{v}\cdot\boldsymbol{u}=\boldsymbol{v}^{\mathrm{T}}\boldsymbol{u}=\begin{bmatrix}3 & 2 & -3\end{bmatrix}\begin{bmatrix}2\\-5\\-1\end{bmatrix}=(3)(2)+(2)(-5)+(-3)(-1)=-1$$

Properties of Inner Product

THEOREM 1

Let $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ be vectors in $\mathbf{R}^n$, and let c be any scalar. Then

a. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$

b. $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$

c. $(c\mathbf{u}) \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (c\mathbf{v})$

d. $\mathbf{u} \cdot \mathbf{u} \geq \mathbf{0}$, and $\mathbf{u} \cdot \mathbf{u} = \mathbf{0}$ if and only if $\mathbf{u} = \mathbf{0}$.

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2. The Length of a Vector（ 长度 ）

For $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$, the **length** or **norm of v** is the nonnegative scalar $\|\mathbf{v}\|$ defined by

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2} \quad \text{and} \quad \|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}.$$

For any scalar c,

$$\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$$

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For example, if $\mathbf{v} = \begin{bmatrix} a \\ b \end{bmatrix}$, then $\|\mathbf{v}\| = \sqrt{a^2 + b^2}$ (distance between **0** and **v**)

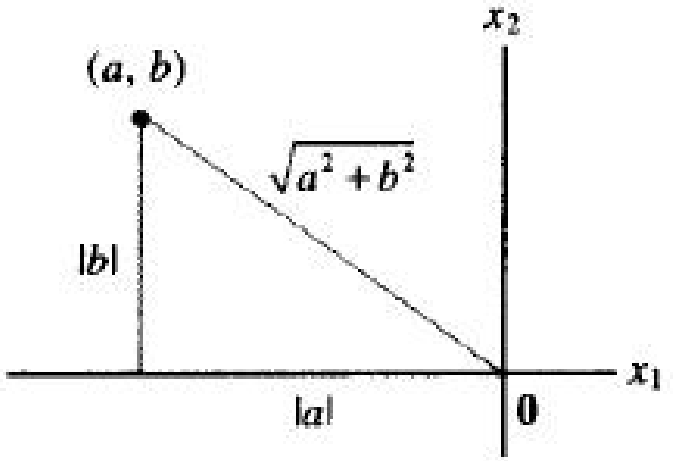


图 6-2 $\|v\|$作为长度的几何意义

Some concepts

- **Unit vector**（单位向量）: A vector whose length is 1 is called a unit vector.
- **Normalizing**（归一化）: If a nonzero vector v is divided by its length (or multiply by $\frac{1}{\|v\|}$), a unit vector u can be obtained. This process is called normalizing.

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Example： Let $\boldsymbol{v} = (1, -2, 2, 0)$, Find a unit vector u in the same direction as v.

Solution： First, compute the length of v:

$$\|\boldsymbol{v}\|^2 = \boldsymbol{v} \cdot \boldsymbol{v} = (1)^2 + (-2)^2 + (2)^2 + (0)^2 = 9 , \quad \|\boldsymbol{v}\| = \sqrt{9} = 3$$

multiply by $\frac{1}{\|\boldsymbol{v}\|}$ to obtain

$$\boldsymbol{u} = \frac{1}{\|\boldsymbol{v}\|}\boldsymbol{v} = \frac{1}{3}\boldsymbol{v} = \frac{1}{3}\begin{bmatrix} 1 \\ -2 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/3 \\ -2/3 \\ 2/3 \\ 0 \end{bmatrix}$$

3. Distance in R^n

The **distance between u and v** in $\mathbf{R}^n$:

$$\text{dist}(\mathbf{u},\mathbf{v}) = \|\mathbf{u}-\mathbf{v}\|.$$

This agrees with the usual formulas for $\mathbf{R}^2$ and $\mathbf{R}^3$. Let $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$.

Then $\mathbf{u} - \mathbf{v} = (u_1 - v_1, u_2 - v_2)$ and

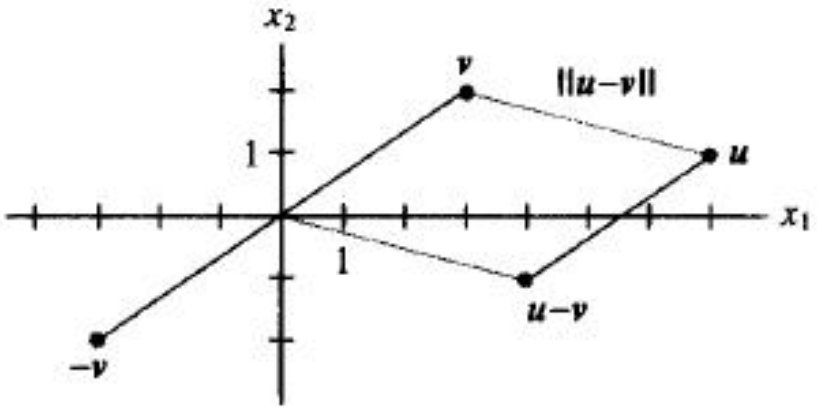
$$\text{dist}(\mathbf{u},\mathbf{v}) = \|\mathbf{u}-\mathbf{v}\| = \|(u_1 - v_1, u_2 - v_2)\|$$

$$= \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2}$$

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Example：Compute the distance between the vectors u=(7,1) and v=(3,2).

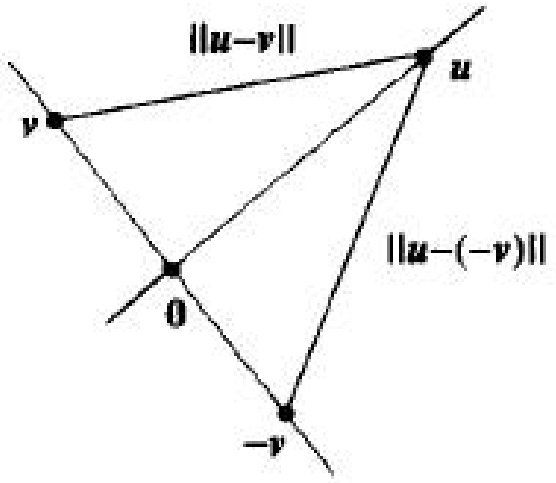
Solution: Compute $u-v=\begin{bmatrix}7\\1\end{bmatrix}-\begin{bmatrix}3\\2\end{bmatrix}=\begin{bmatrix}4\\-1\end{bmatrix}$, So $\|u-v\|=\sqrt{4^2+(-1)^2}=\sqrt{17}$.



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4. Orthogonal Vector（正交向量）

- Consider R^2 or R^3 and two lines through the origin determined by vectors u and v. The two lines are geometrically perpendicular (几何上的垂直) if and only if the distance from u to v is the same as the distance from u to –v.



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$$[\text{dist}(\mathbf{u},\mathbf{v})]^2 = \|\mathbf{u}-\mathbf{v}\|^2 = (\mathbf{u}-\mathbf{v})\cdot(\mathbf{u}-\mathbf{v})$$

$$= (\mathbf{u})\cdot(\mathbf{u}-\mathbf{v}) + (-\mathbf{v})\cdot(\mathbf{u}-\mathbf{v}) =$$

$$= \mathbf{u}\cdot\mathbf{u} - \mathbf{u}\cdot\mathbf{v} + -\mathbf{v}\cdot\mathbf{u} + \mathbf{v}\cdot\mathbf{v}$$

$$= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\mathbf{u}\cdot\mathbf{v}$$

$$\Rightarrow \qquad [\text{dist}(\mathbf{u},\mathbf{v})]^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\mathbf{u}\cdot\mathbf{v}$$

Similarly,

$$[\text{dist}(\mathbf{u},-\mathbf{v})]^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\mathbf{u}\cdot\mathbf{v}$$

Since $[\text{dist}(\mathbf{u}, -\mathbf{v})]^2 = [\text{dist}(\mathbf{u}, \mathbf{v})]^2$, $\mathbf{u} \cdot \mathbf{v} =$ 0.

Two vectors **u** and **v** are said to be **orthogonal** (to each other) if $\mathbf{u} \cdot \mathbf{v} = \mathbf{0}$.

Also note that if **u** and **v** are orthogonal, then
$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

THEOREM 2 THE PYTHAGOREAN THEOREM

Two vectors **u** and **v** are orthogonal if and only if
$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

5. Orthogonal Complements（ 正交补 ）

If a vector **z** is orthogonal to every vector in a subspace W of $\mathbf{R}^n$, then **z** is said to be **orthogonal to** W. The set of vectors **z** that are orthogonal to W is called the **orthogonal complement** of W and is denoted by $W^{\perp}$ (read as "W perp").

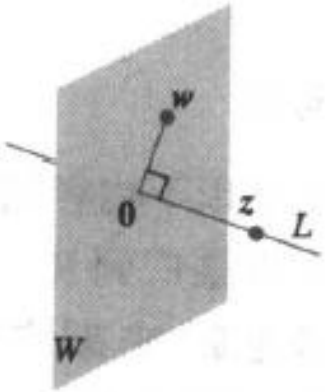


图 6-8　一个作为正交补空间的平面和通过原点的直线

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Two facts about $W^{\perp}$

1. A vector **x** is in $W^{\perp}$ if and only if **x** is orthogonal to every vector in a set that spans W.
2. $W^{\perp}$ is a subspace of R^n.

THEOREM 3

Let A be an $m \times n$ matrix. Then the orthogonal complement of the row space of A is the nullspace of A, and the orthogonal complement of the column space of A is the nullspace of A^T:

$$(\text{Row } A)^{\perp} = \text{Nul } A, \qquad (\text{Col } A)^{\perp} = \text{Nul } A^T.$$

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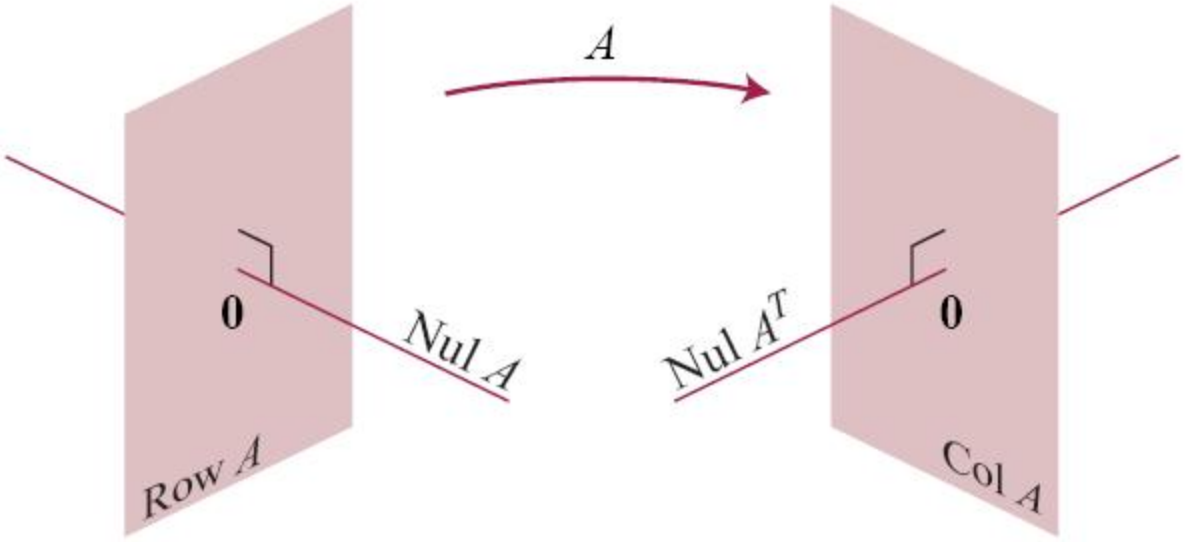


FIGURE 8 The fundamental subspaces of an $m \times n$ matrix A.

Let A be an $m \times n$ matrix. Then the orthogonal complement of the row space of A is the nullspace of A, and the orthogonal complement of the column space of A is the nullspace of A^T:

$$(\text{Row } A)^{\perp} = \text{Nul } A, \qquad (\text{Col } A)^{\perp} = \text{Nul } A^T$$

Assignment

Ex6.1: 11,25,27,31

课堂练习，Ex6.1：19,20

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§6.2 Orthogonal Sets

- 1. Orthogonal Sets (正交集)
- 2. An Orthogonal Projection(正交投影)
- 3. Orthonormal Sets (单位正交集)

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1. Orthogonal Sets (正交集)

Definition :

A set of vectors $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ in $\mathbf{R}^n$ is called an **orthogonal set** if $\mathbf{u}_i \cdot \mathbf{u}_j = 0$ whenever $i \neq j$.

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EXAMPLE: Is $\left\{ \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix} \right\}$ an orthogonal set?

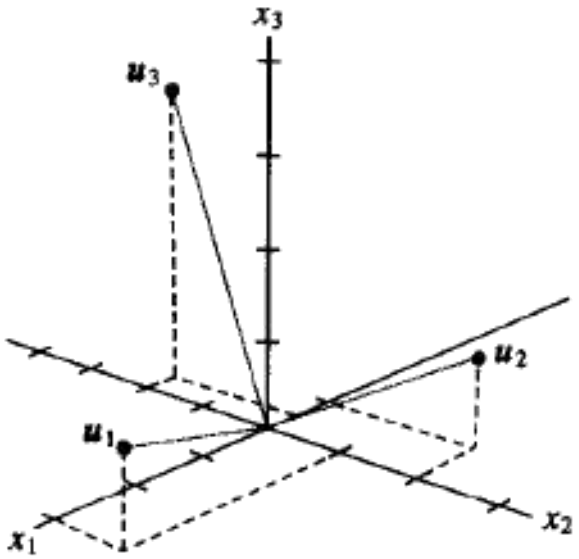
Solution: Label the vectors $\mathbf{u}_1, \mathbf{u}_2,$ and $\mathbf{u}_3$ respectively. Then

$$\boldsymbol{u}_1 \cdot \boldsymbol{u}_2 = 3(-1) + 1(2) + 1(1) = 0$$

$$\boldsymbol{u}_1 \cdot \boldsymbol{u}_3 = 3(-1/2) + 1(-2) + 1(7/2) = 0$$

$$\boldsymbol{u}_2 \cdot \boldsymbol{u}_3 = -1(-1/2) + 2(-2) + 1(7/2) = 0$$

Therefore, $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal set.



正交与线性无关的关系

THEOREM 4

Suppose $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ is an orthogonal set of nonzero vectors in $\mathbf{R}^n$ and W =span$\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$. Then S is a linearly independent set and is therefore a basis for W.

正交必定线性无关（ 不含零向量 ）

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Partial Proof: Suppose

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_p\mathbf{u}_p = \mathbf{0}$$

$$(c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_p\mathbf{u}_p) \cdot \mathbf{u}_1 = \mathbf{0} \cdot$$

$$(c_1\mathbf{u}_1) \cdot \mathbf{u}_1 + (c_2\mathbf{u}_2) \cdot \mathbf{u}_1 + \cdots + (c_p\mathbf{u}_p) \cdot \mathbf{u}_1 = \mathbf{0}$$

$$c_1(\mathbf{u}_1 \cdot \mathbf{u}_1) + c_2(\mathbf{u}_2 \cdot \mathbf{u}_1) + \cdots + c_p(\mathbf{u}_p \cdot \mathbf{u}_1) = \mathbf{0}$$

$$\rightarrow \quad c_1(\mathbf{u}_1 \cdot \mathbf{u}_1) = \mathbf{0}$$

Since $\mathbf{u}_1 \neq \mathbf{0}$, $\mathbf{u}_1 \cdot \mathbf{u}_1 > 0$ which means $c_1 = 0$.

In a similar manner, $c_2, \ldots, c_p$ can be shown to by all 0. So S is a linearly independent set. ■

Orthogonal basis

Definition：

An **orthogonal basis** for a subspace W of $\mathbf{R}^n$ is a basis for W that is also an orthogonal set.

THEOREM 5

Let $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbf{R}^n$. Then each $\mathbf{y}$ in W has a unique representation as a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p$. In fact, if

$$\mathbf{y} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_p\mathbf{u}_p$$

then

$$c_j = \frac{\mathbf{y}\cdot\mathbf{u}_j}{\mathbf{u}_j\cdot\mathbf{u}_j} \qquad (j = 1, \ldots, p)$$

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Proof:

$$\mathbf{y} \cdot \quad = (c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_p\mathbf{u}_p) \cdot$$

$$\mathbf{y} \cdot \mathbf{u}_1 = (c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_p\mathbf{u}_p) \cdot \mathbf{u}_1$$

$$\mathbf{y} \cdot \mathbf{u}_1 = c_1(\mathbf{u}_1 \cdot \mathbf{u}_1) + c_2(\mathbf{u}_2 \cdot \mathbf{u}_1) + \cdots + c_p(\mathbf{u}_p \cdot \mathbf{u}_1)$$

$$\mathbf{y} \cdot \mathbf{u}_1 = c_1(\mathbf{u}_1 \cdot \mathbf{u}_1)$$

$$c_1 = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}$$

Similarly, $c_2 = \dfrac{\underline{y} \cdot \underline{u}_2}{\underline{u}_2 \cdot \underline{u}_2}$, $c_3 = \dfrac{\underline{y} \cdot \underline{u}_3}{\underline{u}_3 \cdot \underline{u}_3}$, …, $c_p = \dfrac{\underline{y} \cdot \underline{u}_p}{\underline{u}_p \cdot \underline{u}_p}$

Example： The set $S=\{u_1, u_2, u_3\}$ is an orthogonal basis for R^3, where

$$u_1=\begin{bmatrix}3\\1\\1\end{bmatrix},\quad u_2=\begin{bmatrix}-1\\2\\1\end{bmatrix},\quad u_3=\begin{bmatrix}-1/2\\-2\\7/2\end{bmatrix}$$

Express the vector $y=\begin{bmatrix}6\\1\\-8\end{bmatrix}$ as a linear combination of the vectors in S.

Solution：Compute

$$y\cdot u_1=11 \qquad y\cdot u_2=-12 \qquad y\cdot u_3=-33$$

$$u_1\cdot u_1=11 \qquad u_2\cdot u_2=6 \qquad u_3\cdot u_3=33/2$$

By Theorem 5,

$$\begin{aligned} y&=\frac{y\cdot u_1}{u_1\cdot u_1}\cdot u_1+\frac{y\cdot u_2}{u_2\cdot u_2}\cdot u_2+\frac{y\cdot u_3}{u_3\cdot u_3}\cdot u_3\\ &=\frac{11}{11}u_1+\frac{-12}{6}u_2+\frac{-33}{33/2}u_3\\ &=u_1-2u_2-2u_3 \end{aligned}$$

2. An Orthogonal Projection

For a nonzero vector **u** in $\mathbf{R}^n$, suppose we want to write **y** in $\mathbf{R}^n$ as the the following

$$\mathbf{y} = (\text{multiple of } \mathbf{u}) + (\text{multiple a vector } \perp \text{ to } \mathbf{u})$$

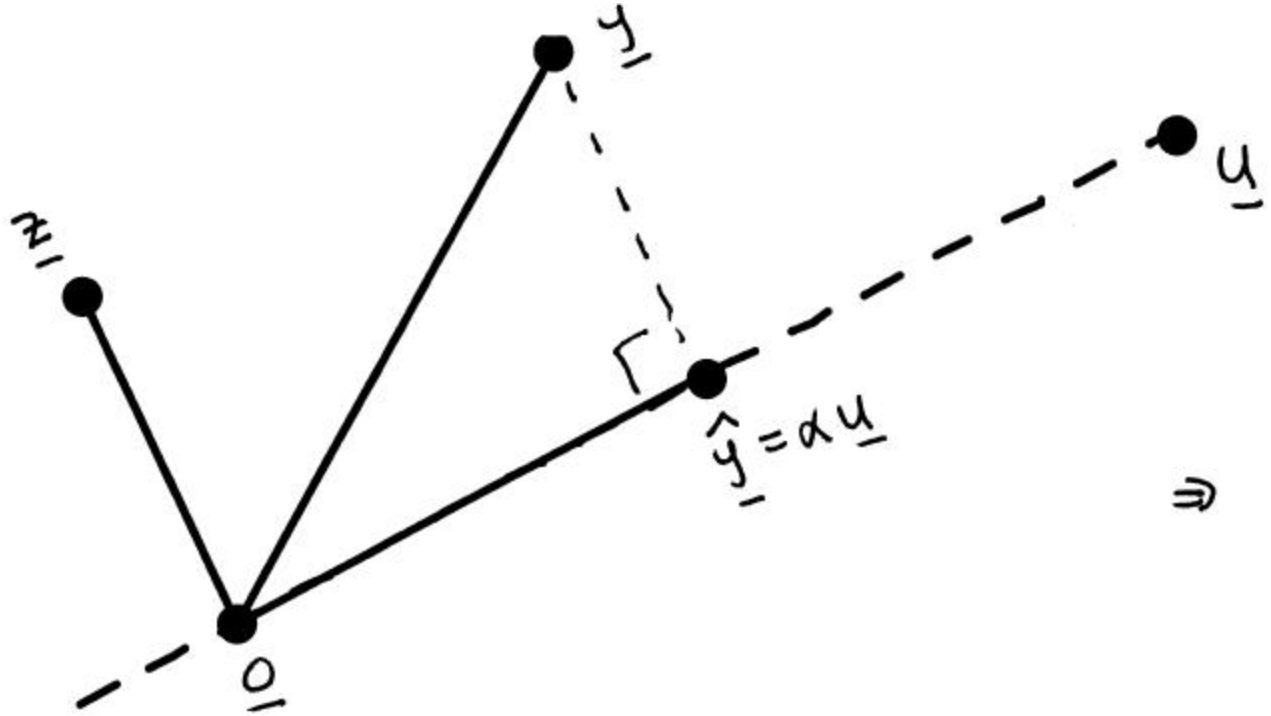
$$(\mathbf{y} - \alpha\mathbf{u}) \cdot \mathbf{u} = 0$$

$$\mathbf{y} \cdot \mathbf{u} - \alpha(\mathbf{u} \cdot \mathbf{u}) = 0 \qquad \Rightarrow \qquad \alpha = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}$$

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u} \qquad \textbf{(orthogonal projection of y onto u)}$$

and

$$\mathbf{z} = \mathbf{y} - \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u} \qquad \textbf{(component of y orthogonal to u)}$$



$$\underline{y} = \alpha \underline{u} + \underline{z}$$

$$\Rightarrow \quad \underline{z} = \underline{y} - \alpha \underline{u}$$

Example : $y=\begin{bmatrix}7\\6\end{bmatrix}$ and $u=\begin{bmatrix}4\\2\end{bmatrix}$. Find the orthogonal projection of y onto u. then write y as the sum of two orthogonal vectors, one in Span {u} and one orthogonal to u.

Solution: Compute

$$y\cdot u=\begin{bmatrix}7\\6\end{bmatrix}\cdot\begin{bmatrix}4\\2\end{bmatrix}=40$$

$$u\cdot u=\begin{bmatrix}4\\2\end{bmatrix}\cdot\begin{bmatrix}4\\2\end{bmatrix}=20$$

The orthogonal projection of y onto u is

$$\hat{y}=\frac{y\cdot u}{u\cdot u}\cdot u=\frac{40}{20}u=2\begin{bmatrix}4\\2\end{bmatrix}=\begin{bmatrix}8\\4\end{bmatrix}$$

And the component of y orthogonal to u is

$$y-\hat{y}=\begin{bmatrix}7\\6\end{bmatrix}-\begin{bmatrix}8\\4\end{bmatrix}=\begin{bmatrix}-1\\2\end{bmatrix}$$

The orthogonal projection of y onto u is

$$\begin{bmatrix}7\\6\end{bmatrix}=\begin{bmatrix}8\\4\end{bmatrix}+\begin{bmatrix}-1\\2\end{bmatrix}$$

y $\hat{y}$ $(y-\hat{y})$

The decomposition of y is illustrated in the following Fig. Note: If the calculations above are correct, then $\{y, y-\hat{y}\}$ will be an orthogonal set. As a check, compute：

$$\hat{y}\cdot(y-\hat{y})=\begin{bmatrix}8\\4\end{bmatrix}\cdot\begin{bmatrix}-1\\2\end{bmatrix}=-8+8=0$$

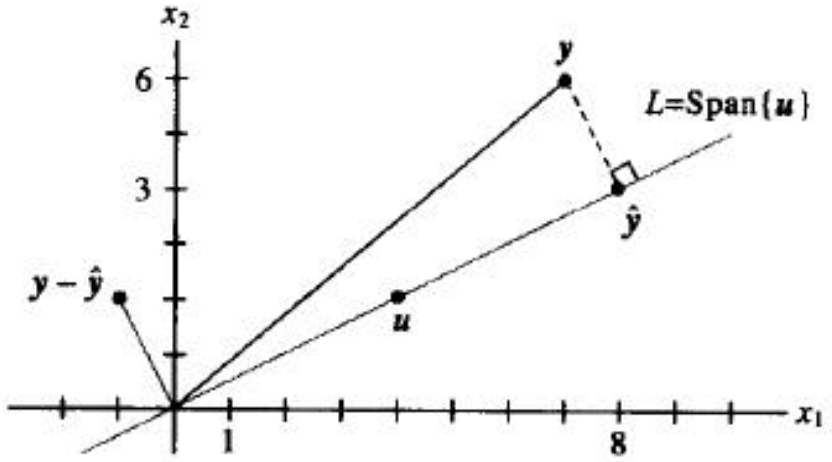


图 6-13　y 在通过原点的直线 L 上的正交投影

3. Orthonormal Sets (单位正交集)

Orthonormal Sets

A set of vectors $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ in $\mathbf{R}^n$ is called an **orthonormal set** if it is an orthogonal set of unit vectors.

If $W = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$, then $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ is an orthonormal basis for W.

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THEOREM 6 An $m \times n$ matrix U has orthonormal columns if and only if $U^TU = I$.

Recall that $\mathbf{v}$ is a unit vector if $\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{\mathbf{v}^T\mathbf{v}} = 1$.

Suppose $U = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{bmatrix}$ where $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthonormal set.

Then $U^TU = \begin{bmatrix} \mathbf{u}_1^T \\ \mathbf{u}_2^T \\ \mathbf{u}_3^T \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{bmatrix} = \begin{bmatrix} \mathbf{u}_1^T\mathbf{u}_1 & \mathbf{u}_1^T\mathbf{u}_2 & \mathbf{u}_1^T\mathbf{u}_3 \\ \mathbf{u}_2^T\mathbf{u}_1 & \mathbf{u}_2^T\mathbf{u}_2 & \mathbf{u}_2^T\mathbf{u}_3 \\ \mathbf{u}_3^T\mathbf{u}_1 & \mathbf{u}_3^T\mathbf{u}_2 & \mathbf{u}_3^T\mathbf{u}_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

If U in Theorem 6 is a square matrix, then we have $U^{-1}=U^T$, and in this case, it is called an orthogonal matrix（ 正交矩阵 ）. Also, it can be verified that $UU^T = I$

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§ 6.2 Orthogonal Sets

THEOREM 7 Let U be an $m \times n$ matrix with orthonormal columns, and let $\mathbf{x}$ and $\mathbf{y}$ be in $\mathbf{R}^n$. Then

a. $\|U\mathbf{x}\| = \|\mathbf{x}\|$

b. $(U\mathbf{x}) \cdot (U\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$

c. $(U\mathbf{x}) \cdot (U\mathbf{y}) = 0$ if and only if $\mathbf{x} \cdot \mathbf{y} = 0$.

Prove b

一类非常有用的矩阵

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Assignment

- Ex6.2: 14,17,27,28

- 课堂练习 Ex6.2: 23,24

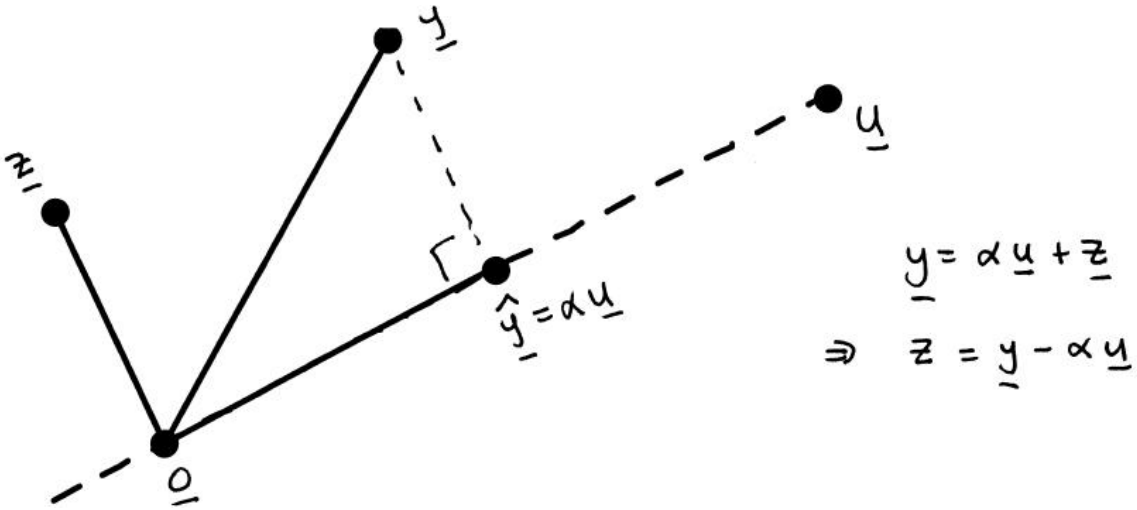
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§6.3 Orthogonal Projections

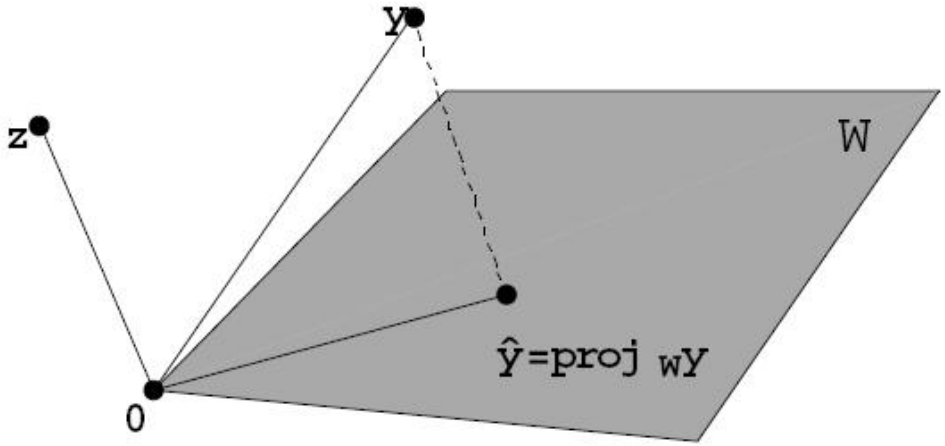
- 1. Orthogonal Projection
- 2. Geometric Interpretation of the Orthogonal Projection
- 3. Properties of Orthogonal Projections

向量到平面的投影

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到直线的投影



到平面的投影

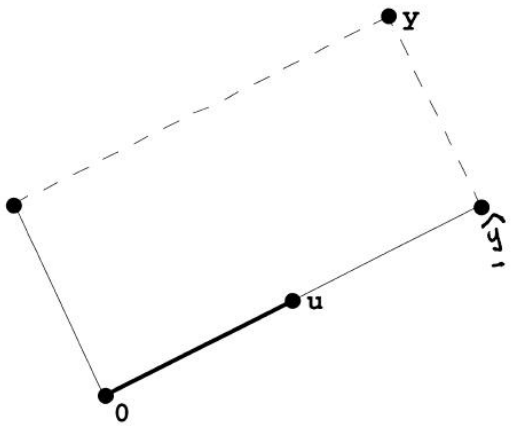
回顾

Review

$\hat{\mathbf{y}} = \frac{\mathbf{y}\cdot\mathbf{u}}{\mathbf{u}\cdot\mathbf{u}}\mathbf{u}$ is the **orthogonal projection of** $\mathbf{y}$ **onto** $\mathbf{u}$.

Suppose $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is an orthogonal basis for W in $\mathbf{R}^n$. For each $\mathbf{y}$ in W,

$$\mathbf{y} = \left(\frac{\mathbf{y}\cdot\mathbf{u}_1}{\mathbf{u}_1\cdot\mathbf{u}_1}\right)\mathbf{u}_1 + \cdots + \left(\frac{\mathbf{y}\cdot\mathbf{u}_p}{\mathbf{u}_p\cdot\mathbf{u}_p}\right)\mathbf{u}_p$$



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从到某条直线的投影到某个平面的投影

THEOREM 8
THE ORTHOGONAL DECOMPOSITION THEOREM

Let W be a subspace of $\mathbf{R}^n$. Then each $\mathbf{y}$ in $\mathbf{R}^n$ can be uniquely represented in the form

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$$

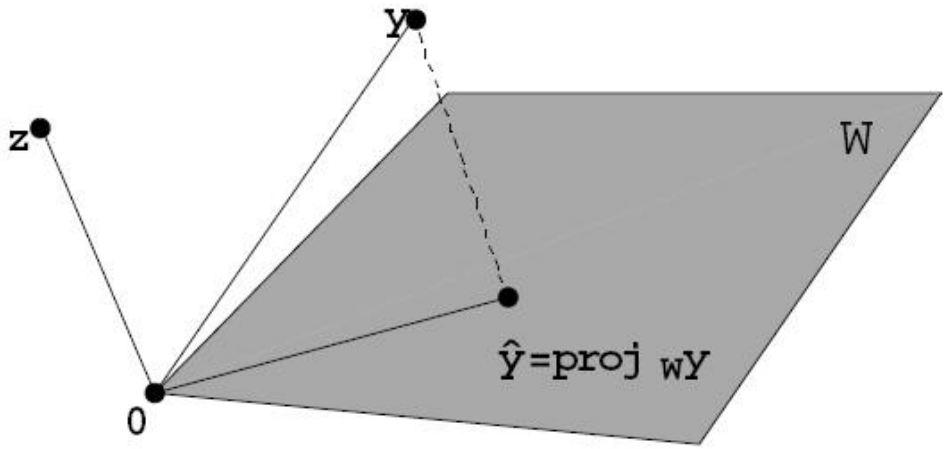
where $\hat{\mathbf{y}}$ is in W and $\mathbf{z}$ is in $W^{\perp}$. In fact, if $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is any orthogonal basis of W, then

$$\hat{\mathbf{y}} = \left(\frac{\mathbf{y}\cdot\mathbf{u}_1}{\mathbf{u}_1\cdot\mathbf{u}_1}\right)\mathbf{u}_1 + \cdots + \left(\frac{\mathbf{y}\cdot\mathbf{u}_p}{\mathbf{u}_p\cdot\mathbf{u}_p}\right)\mathbf{u}_p$$

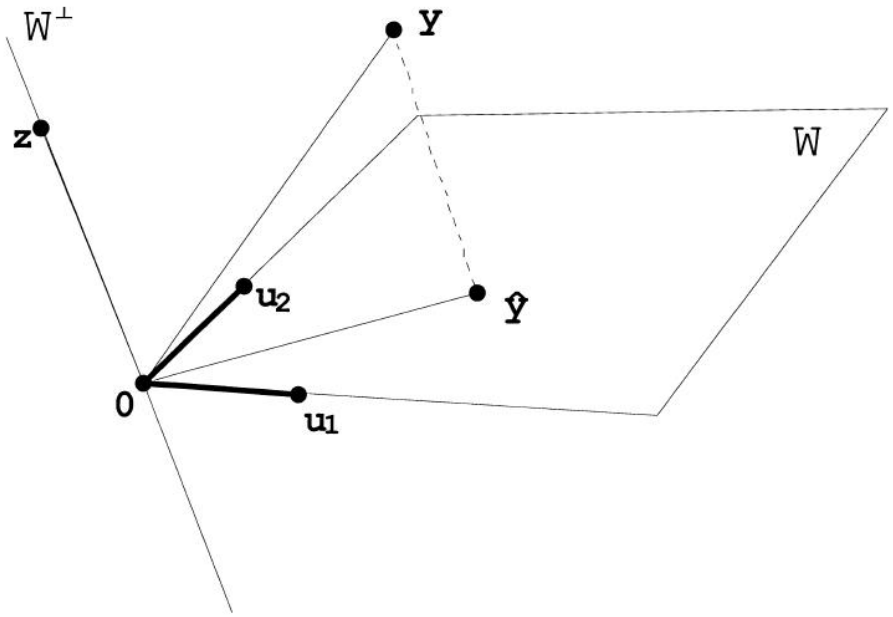
and $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$.

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The vector $\hat{\mathbf{y}}$ is called the **orthogonal projection of y onto** W.



EXAMPLE: Suppose $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal basis for $\mathbf{R}^3$ and let $W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$. Write $\mathbf{y}$ in $\mathbf{R}^3$ as the sum of a vector $\hat{\mathbf{y}}$ in W and a vector $\mathbf{z}$ in $W^\perp$.



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Solution: Write

$$\mathbf{y} = \left(\frac{\mathbf{y}\cdot\mathbf{u}_1}{\mathbf{u}_1\cdot\mathbf{u}_1}\right)\mathbf{u}_1 + \left(\frac{\mathbf{y}\cdot\mathbf{u}_2}{\mathbf{u}_2\cdot\mathbf{u}_2}\right)\mathbf{u}_2 + \left(\frac{\mathbf{y}\cdot\mathbf{u}_3}{\mathbf{u}_3\cdot\mathbf{u}_3}\right)\mathbf{u}_3$$

where

$$\hat{\mathbf{y}} = \left(\frac{\mathbf{y}\cdot\mathbf{u}_1}{\mathbf{u}_1\cdot\mathbf{u}_1}\right)\mathbf{u}_1 + \left(\frac{\mathbf{y}\cdot\mathbf{u}_2}{\mathbf{u}_2\cdot\mathbf{u}_2}\right)\mathbf{u}_2$$

$$\mathbf{z} = \left(\frac{\mathbf{y}\cdot\mathbf{u}_3}{\mathbf{u}_3\cdot\mathbf{u}_3}\right)\mathbf{u}_3.$$

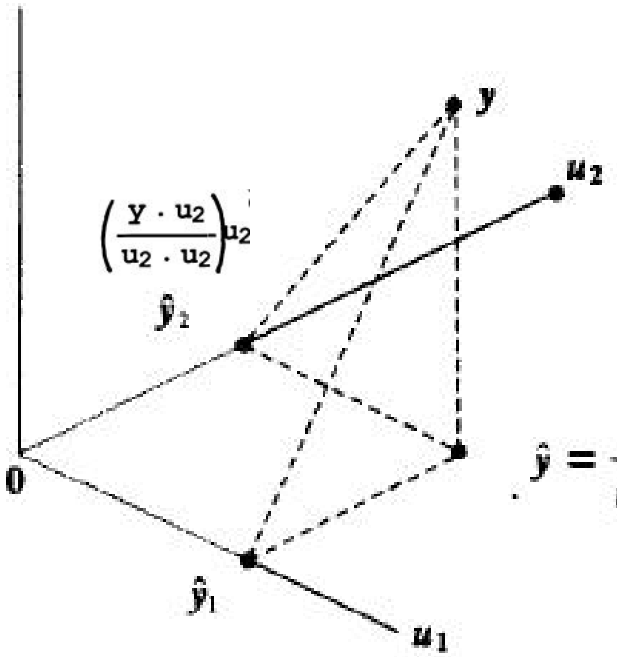
To show that $\mathbf{z}$ is orthogonal to every vector in W, show that $\mathbf{z}$ is orthogonal to the vectors in $\{\mathbf{u}_1, \mathbf{u}_2\}$.

$$\mathbf{z}\cdot\mathbf{u}_1 = \left(\frac{\mathbf{y}\cdot\mathbf{u}_3}{\mathbf{u}_3\cdot\mathbf{u}_3}\mathbf{u}_3\right)\mathbf{u}_1 = \left(\frac{\mathbf{y}\cdot\mathbf{u}_3}{\mathbf{u}_3\cdot\mathbf{u}_3}\right)\mathbf{u}_3\,\mathbf{u}_1 = \mathbf{0}$$

$$\mathbf{z}\cdot\mathbf{u}_2 = \left(\frac{\mathbf{y}\cdot\mathbf{u}_3}{\mathbf{u}_3\cdot\mathbf{u}_3}\mathbf{u}_3\right)\mathbf{u}_2 = \left(\frac{\mathbf{y}\cdot\mathbf{u}_3}{\mathbf{u}_3\cdot\mathbf{u}_3}\right)\mathbf{u}_3\,\mathbf{u}_2 = \mathbf{0}$$

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几何解释



$$\hat{y} = \frac{y \cdot u_1}{u_1 \cdot u_1} u_1 + \frac{y \cdot u_2}{u_2 \cdot u_2} u_2 = \hat{y}_1 + \hat{y}_2$$

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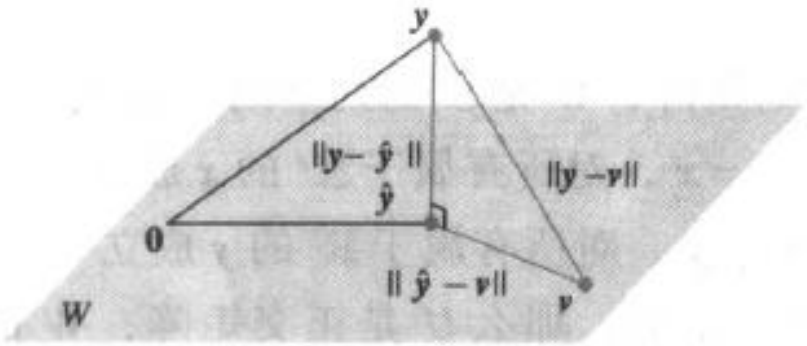
点到平面的距离

THEOREM 9 The Best Approximation Theorem

Let W be a subspace of $\mathbf{R}^n$, $\mathbf{y}$ any vector in $\mathbf{R}^n$, and $\hat{\mathbf{y}}$ the orthogonal projection of $\mathbf{y}$ onto W. Then $\hat{\mathbf{y}}$ is the point in W closest to $\mathbf{y}$, in the sense that

$$\|\mathbf{y} - \hat{\mathbf{y}}\| < \|\mathbf{y} - \mathbf{v}\|$$

for all $\mathbf{v}$ in W distinct from $\hat{\mathbf{y}}$.



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Outline of Proof: Let $\mathbf{v}$ in W distinct from $\hat{\mathbf{y}}$. Then

$\mathbf{v} - \hat{\mathbf{y}}$ is also in W (why?)

$\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to $W \quad \Rightarrow \quad \mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to $\mathbf{v} - \hat{\mathbf{y}}$

$$\mathbf{y} - \mathbf{v} = (\mathbf{y} - \hat{\mathbf{y}}) + (\hat{\mathbf{y}} - \mathbf{v}) \qquad \Rightarrow \qquad \|\mathbf{y} - \mathbf{v}\|^2 = \|\mathbf{y} - \hat{\mathbf{y}}\|^2 + \|\hat{\mathbf{y}} - \mathbf{v}\|^2.$$

$$\|\mathbf{y} - \mathbf{v}\|^2 > \|\mathbf{y} - \hat{\mathbf{y}}\|^2$$

Hence, $\|\mathbf{y} - \hat{\mathbf{y}}\| < \|\mathbf{y} - \mathbf{v}\|$. ■

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EXAMPLE: Find the closest point to $\mathbf{y}$ in Span$\{\mathbf{u}_1, \mathbf{u}_2\}$ where

$$\mathbf{y} = \begin{bmatrix} 2 \\ 4 \\ 0 \\ -2 \end{bmatrix}, \mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \text{ and } \mathbf{u}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}.$$

Solution: $\hat{\mathbf{y}} = \left(\frac{\mathbf{y}\cdot\mathbf{u}_1}{\mathbf{u}_1\cdot\mathbf{u}_1}\right)\mathbf{u}_1 + \left(\frac{\mathbf{y}\cdot\mathbf{u}_2}{\mathbf{u}_2\cdot\mathbf{u}_2}\right)\mathbf{u}_2$

$$= (\ 3\ ) \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + (\ -1\ ) \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ -1 \\ -1 \end{bmatrix}$$

当平面有标准正交基的时候

THEOREM 10

If $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is an orthonormal basis for a subspace W of $\mathbf{R}^n$, then

$$\text{proj}_W \mathbf{y} = (\mathbf{y} \cdot \mathbf{u}_1)\mathbf{u}_1 + \cdots + (\mathbf{y} \cdot \mathbf{u}_p)\mathbf{u}_p$$

If $U = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_p \end{bmatrix}$, then

$$\text{proj}_W \mathbf{y} = UU^T\mathbf{y} \quad \text{for all } \mathbf{y} \text{ in } \mathbf{R}^n.$$

Outline of Proof:

$$\text{proj}_W \mathbf{y} = \left(\frac{\mathbf{y}\cdot\mathbf{u}_1}{\mathbf{u}_1\cdot\mathbf{u}_1}\right)\mathbf{u}_1 + \cdots + \left(\frac{\mathbf{y}\cdot\mathbf{u}_p}{\mathbf{u}_p\cdot\mathbf{u}_p}\right)\mathbf{u}_p$$

$$= (\mathbf{y} \cdot \mathbf{u}_1)\mathbf{u}_1 + \cdots + (\mathbf{y} \cdot \mathbf{u}_p)\mathbf{u}_p = UU^T\mathbf{y}.$$

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Assignment

- Ex6.3:11,24

- 课堂练习 Ex6.3:21,22

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正交投影：当平面有标准正交基的时候

THEOREM 10

If $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is an orthonormal basis for a subspace W of $\mathbf{R}^n$, then

$$\text{proj}_W \mathbf{y} = (\mathbf{y} \cdot \mathbf{u}_1)\mathbf{u}_1 + \cdots + (\mathbf{y} \cdot \mathbf{u}_p)\mathbf{u}_p$$

If $U = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_p \end{bmatrix}$, then

$$\text{proj}_W \mathbf{y} = UU^T\mathbf{y} \quad \text{for all } \mathbf{y} \text{ in } \mathbf{R}^n.$$

Outline of Proof:

$$\text{proj}_W \mathbf{y} = \left(\frac{\mathbf{y}\cdot\mathbf{u}_1}{\mathbf{u}_1\cdot\mathbf{u}_1}\right)\mathbf{u}_1 + \cdots + \left(\frac{\mathbf{y}\cdot\mathbf{u}_p}{\mathbf{u}_p\cdot\mathbf{u}_p}\right)\mathbf{u}_p$$

$$= (\mathbf{y} \cdot \mathbf{u}_1)\mathbf{u}_1 + \cdots + (\mathbf{y} \cdot \mathbf{u}_p)\mathbf{u}_p = UU^T\mathbf{y}.$$

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§ 6.4 The Gram – Schmidt Process

- **What is Gram – Schmidt Process?**

目标：找到给定空间W的一组正交基

前提：已知该空间的一组非正交基

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EXAMPLE: Suppose $W = \text{Span}\{\mathbf{x}_1, \mathbf{x}_2\}$ where $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\mathbf{x}_2 = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$. Find an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ for W.

Let

$$\mathbf{v}_1 = \mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

$$\hat{\mathbf{y}} = \text{proj}_{\mathbf{v}_1}\mathbf{x}_2 = \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1$$

and

$$\mathbf{v}_2 = \mathbf{x}_2 - \hat{\mathbf{y}} = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} - \frac{4}{2}\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$$

(component of $\mathbf{x}_2$ orthogonal to $\mathbf{x}_1$)

So $\{v_1, v_2\}$ is an orthogonal basis for W.

EXAMPLE: Suppose $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ is a basis for a subspace W of $\mathbf{R}^4$. Describe an orthogonal basis for W.

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THEOREM 11 THE GRAM-SCHMIDT PROCESS

Given a basis $\{\mathbf{x}_1, \ldots, \mathbf{x}_p\}$ for a subspace W of $\mathbf{R}^n$, define

$$\mathbf{v}_1 = \mathbf{x}_1$$

$$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1$$

$$\mathbf{v}_3 = \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2$$

$$\vdots$$

$$\mathbf{v}_p = \mathbf{x}_p - \frac{\mathbf{x}_p \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{x}_p \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2 - \cdots - \frac{\mathbf{x}_p \cdot \mathbf{v}_{p-1}}{\mathbf{v}_{p-1} \cdot \mathbf{v}_{p-1}}\mathbf{v}_{p-1}$$

The $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is an orthogonal basis for W and

$$\text{Span}\{x_1,\ldots,x_k\} = \text{Span}\{v_1,\ldots,v_k\}$$

for $1 \le k \le p$.

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归纳法 (证Span{$x_1,\ldots,x_k$}= Span{$v_1,\ldots,v_k$}).

Proof ： For 1≤ k ≤p, let W_k = Span{$x_1,\ldots,x_k$}.

Basis: Set $v_1 = x_1$, so that span{v_1} = Span{x_1}.

Induction: Suppose, for some k< p, we have constructed $v_1,\ldots,v_k$ so that {$v_1,\ldots,v_k$} is an orthogonal basis for W_k.

Define

$$\boldsymbol{v}_{k+1} = \boldsymbol{x}_{k+1} - \operatorname{proj}_{W_k} \boldsymbol{x}_{k+1}$$

Then v_{k+1} is orthogonal to W_k. Further more,v_{k+1}≠ 0 because x_{k+1} is not in W_k = Span{$x_1,\ldots,x_k$}.

Hence {$v_1,\ldots,v_{k+1}$} is an orthogonal set of nonzero vectors in the (k+1)-dimension space W_{k+1}.

By the basis theorem in Section 4.5, this set is an orthogonal basis for W_{k+1}.

Hence W_{k+1} = Span{$v_1,\ldots,v_{k+1}$}.

§ 6.4 The Gram – Schmidt Process

EXAMPLE Suppose $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$, where

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}, \mathbf{x}_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix}, \mathbf{x}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix},$$

is a basis for a subspace W of $\mathbf{R}^4$. Describe an orthogonal basis for W.

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Solution: $\mathbf{v}_1 = \mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}$ and

$$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix} - \frac{5}{14}\begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{9}{14} \\ \frac{9}{7} \\ -\frac{15}{14} \\ 0 \end{bmatrix}$$

Replace $\mathbf{v}_2$ with $14\mathbf{v}_2$: $\mathbf{v}_2 = 14\begin{bmatrix} \frac{9}{14} \\ \frac{9}{7} \\ -\frac{15}{14} \\ 0 \end{bmatrix} = \begin{bmatrix} 9 \\ 18 \\ -15 \\ 0 \end{bmatrix}$

(optional step - to make $\mathbf{v}_2$ easier to work with in the next step)

$$\mathbf{v}_3 = \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2$$

$$\mathbf{v}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{14}\begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix} - \frac{9}{630}\begin{bmatrix} 9 \\ 18 \\ -15 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{14}\begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix} - \frac{1}{70}\begin{bmatrix} 9 \\ 18 \\ -15 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{4}{5} \\ -\frac{2}{5} \\ 0 \\ 1 \end{bmatrix}$$

Rescale (optional): $\mathbf{v}_3 = \begin{bmatrix} 4 \\ -2 \\ 0 \\ 5 \end{bmatrix}$

Orthogonal Basis for W:

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 9 \\ 18 \\ -15 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ -2 \\ 0 \\ 5 \end{bmatrix} \right\}$$

QR Factorization of Matrices

THEOREM 12 (The QR Factorization)

If A is an $m \times n$ matrix with linearly independent columns, then A can be factored as $A = QR$, where Q is an $m \times n$ matrix whose columns form an orthonormal basis for Col A and R is an $n \times n$ upper triangular invertible matrix with positive entries on its main diagonal.

可以看做是Gram-schmidt过程的推论

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- Proof : The columns of A form a basis $\{x_1,\ldots,x_n\}$ for Col A. Construct an orthonormal basis $\{u_1,\ldots,u_n\}$ for W = Col A.

$$\text{Span}\{x_1,\cdots,x_k\} = \text{Span}\{u_1,\cdots,u_k\}.$$

Let $Q = [u_1 \;\; u_2 \;\; \cdots \;\; u_n]$

For k = 1,…,n, x_k can be written as

$$x_k = r_{1k}u_1 + \cdots + r_{kk}u_k + 0\cdot u_{k+1} + \cdots + 0\cdot u_n$$

Assume that $r_{kk} \geqslant 0$ (Why?).

This shows that x_k is a linear combination of the columns of Q using the entries in the vector:

$$r_k = \begin{bmatrix} r_{1k} \\ \vdots \\ r_{kk} \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

That is $x_k = Qr_k$, $k = 1,\cdots,n$,

Let $R = [r_1 \cdots r_n]$. Then $A = [x_1 \cdots x_n] = [Qr_1 \cdots Qr_n] = QR$

EXAMPLE Find the QR factorization of $A = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 3 \end{bmatrix}$.

Solution: Use the Gram Schmidt process to find

$\left\{ \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ which is an orthonomal basis for

$\text{col } A = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} \right\}$. So $Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 0 \\ 0 & 1 \end{bmatrix}$.

Since Q has orthonormal columns, $Q^TQ = I$. So if $A = QR$, then

$$\underline{\quad}A = \underline{\quad}QR$$

$$R = Q^TA = \begin{bmatrix} \frac{1}{2}\sqrt{2} & \frac{1}{2}\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} \sqrt{2} & 2\sqrt{2} \\ 0 & 3 \end{bmatrix}$$

Assignment

- Ex6.4: 9, 13

- 课堂练习 Ex6.4:17,18

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§6.5 Least – Squares Problems(最小二乘问题)

- **Problem:** What do we do when $A\mathbf{x} = \mathbf{b}$ has no solution $\mathbf{x}$?

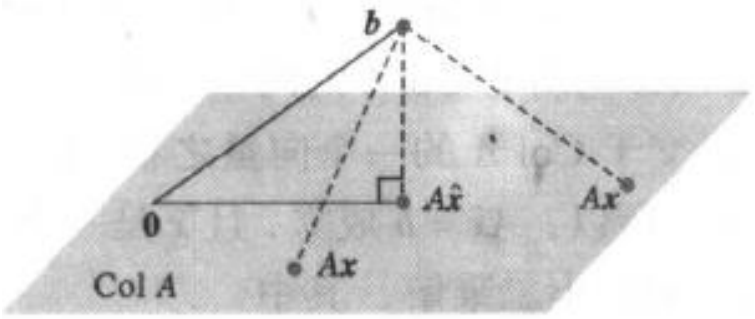
Answer: Find $\hat{\mathbf{x}}$ such that $A\hat{\mathbf{x}}$ is as "close" as possible to $\mathbf{b}$. (*Least Squares Problem*)

关键：理解几何意义

If A is $m \times n$ and $\mathbf{b}$ is in $\mathbf{R}^m$, a **least-squares solution** of $A\mathbf{x} = \mathbf{b}$ is an $\hat{\mathbf{x}}$ in $\mathbf{R}^n$ such that

$$\|\mathbf{b} - A\hat{\mathbf{x}}\| \leq \|\mathbf{b} - A\mathbf{x}\|$$

for all $\mathbf{x}$ in $\mathbf{R}^n$.

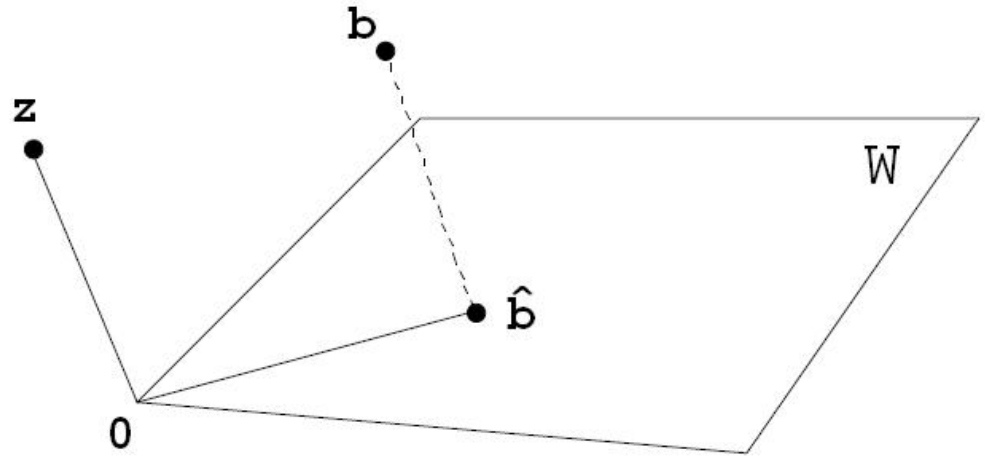


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Solution of the General Least-Squares Problem

Let $W = \text{Col } A$ where A is $m \times n$ and $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$.

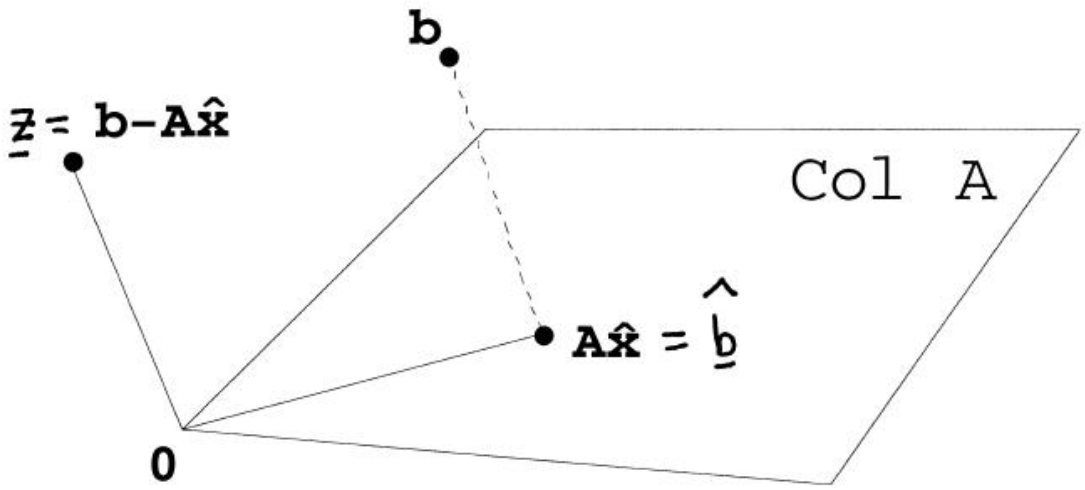
Suppose $\mathbf{b}$ is in $\mathbf{R}^m$ and $\hat{\mathbf{b}} = \text{proj}_W \mathbf{b}$.



$\hat{\mathbf{b}}$ is the point in $W = \text{Col } A$ closest to $\mathbf{b}$

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Since $\hat{\mathbf{b}}$ is in Col A, then $\hat{\mathbf{x}}$ is a vector in $\mathbf{R}^n$ such that $\hat{\mathbf{b}} = A\hat{\mathbf{x}}$.



By the Orthogonal Projection Theorem, $\mathbf{z}$ is in $W^{\perp}$ where $\mathbf{z} = \mathbf{b} - A\hat{\mathbf{x}}$.

Then $\mathbf{b} - A\hat{\mathbf{x}}$ is orthogonal to every column of A. Meaning that

$$\mathbf{a}_1^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0 \qquad \mathbf{a}_2^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0 \qquad \cdots \qquad \mathbf{a}_n^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0$$

$$\begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_n^T \end{bmatrix} (\mathbf{b} - A\hat{\mathbf{x}}) = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$$

$$A^T\mathbf{b} - A^TA\hat{\mathbf{x}} = \mathbf{0}$$

$$\boxed{A^TA\hat{\mathbf{x}} = A^T\mathbf{b}}$$

(normal equations for $\hat{\mathbf{x}}$)

THEOREM 13

The set of least squares solutions of $A\mathbf{x} = \mathbf{b}$ is the set of all solutions of the normal equations $A^TA\hat{\mathbf{x}} = A^T\mathbf{b}$.

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EXAMPLE: Find a least squares solution to the inconsistent system $A\mathbf{x} = \mathbf{b}$ where

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 2 & 2 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Solution: Solve $A^TA\hat{\mathbf{x}} = A^T\mathbf{b}$ after first finding A^TA and $A^T\mathbf{b}$.

$$A^TA = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 1 & 2 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 5 \\ 4 & 5 \end{bmatrix}$$

$$A^T\mathbf{b} = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 8 \\ 8 \end{bmatrix}$$

So solve the following:

$$\underbrace{\begin{bmatrix} 8 & 5 \\ 4 & 5 \end{bmatrix}}_{A^TA}\underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\hat{\mathbf{x}}} = \underbrace{\begin{bmatrix} 8 \\ 8 \end{bmatrix}}_{A^T\mathbf{b}}$$

$$\begin{bmatrix} 8 & 5 & 8 \\ 4 & 5 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 5/8 \end{bmatrix}$$

$$\Rightarrow \qquad \hat{\mathbf{x}} = \begin{bmatrix} 0 \\ 5/8 \end{bmatrix}$$

When A^TA is invertible,

$$A^TA\hat{\mathbf{x}} = A^T\mathbf{b}$$

$$(A^TA)^{-1}A^TA\hat{\mathbf{x}} = (A^TA)^{-1}A^T\mathbf{b}$$

$$\hat{\mathbf{x}} = (A^TA)^{-1}A^T\mathbf{b}$$

So in the last example,

$$(A^TA)^{-1} = \begin{bmatrix} 8 & 5 \\ 4 & 5 \end{bmatrix}^{-1} = \begin{bmatrix} 1/4 & -1/4 \\ 1/5 & 2/5 \end{bmatrix}$$

$$\hat{\mathbf{x}} = (A^TA)^{-1}A^T\mathbf{b} = \begin{bmatrix} 1/4 & -1/4 \\ 1/5 & 2/5 \end{bmatrix} \begin{bmatrix} 8 \\ 8 \end{bmatrix} = \hat{\mathbf{x}} = \begin{bmatrix} 0 \\ 5/8 \end{bmatrix}$$

THEOREM 14

The matrix A^TA is invertible if and only if the columns of A are linearly independent. In this case, the equation $A\mathbf{x} = \mathbf{b}$ has only one least-squares solution $\hat{\mathbf{x}}$, and it is given by

$$\hat{\mathbf{x}} = (A^TA)^{-1}A^T\mathbf{b}.$$

证明过程见Ex6.5中的习题19，20,21

$$\text{least-squares error} = \|\mathbf{b} - A\hat{\mathbf{x}}\|$$

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- **Alternative Calculations of Least-Squares Solutions**

其他的计算方法

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The column vectors of A are linearly independent

Theorem 15

Given an m*n matrix A with linearly independent columns, let A = QR be a QR factorization of A as in Theorem 12. Then, for each b in R^m, the equation Ax = b has a unique least-squares solution, given by

$$\hat{x} = R^{-1}Q^{T}b$$

Proof： Let $\hat{x} = R^{-1}Q^{T}b$, then

$$A\hat{x} = QR\hat{x} = QRR^{-1}Q^{T}b = QQ^{T}b .$$

$QQ^{T}b$ is the orthogonal projection $\hat{b}$ of b onto Col A. This shows that $\hat{x}$ is a least-squares solution of Ax = b.

The uniqueness followings from Theorem 14.

Assignment

- Ex6.5: 2,19,22

- 课堂练习 17,18

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6.7 Inner product Space（ 内积空间 ）

- 内积
- 内积空间
- 柯西-许瓦茨不等式
- 三角不等式

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